

EMBEDDED LINUX SYSTEMS

LAB-03 REPORT

Debugging & Application Performance Analysis

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Lab	LAB-03 — Embedded Linux Systems
Execution date	09 October 2026
Host kernel	6.8.0-138-generic
GDB	GNU gdb 12.1
Valgrind	3.18.1
perf	6.8.12
GCC	11.4.0

Prepared from the available LAB03 source files and recorded terminal outputs.

1. Objectives

The lab investigates debugging and performance analysis workflows for embedded Linux applications: GDB debugging, memory-leak detection with Valgrind, system-call analysis with strace, CPU profiling with perf, and latency measurement with cyclicttest. The target environment used in the practical work includes an ARMv7-A Cortex-A9 QEMU vexpress-a9 system and an initramfs reused from earlier labs.

2. Environment

Item	Recorded value
Host kernel	6.8.0-138-generic
GDB	GNU gdb (Ubuntu 12.1-0ubuntu1~22.04.2) 12.1
Valgrind	valgrind-3.18.1
perf	perf version 6.8.12
GCC	gcc (Ubuntu 11.4.0-1ubuntu1~22.04.3) 11.4.0
Target kernel	Linux 5.15.0 ARMv7 (QEMU boot log / previous validation)
Target machine	QEMU vexpress-a9

3. Section A — GDB Debugging

The available log file (qemu_gdb.log) records demonstrations involving an array out-of-bounds/invalid value, a use-after-free case, and signed integer overflow. The excerpt at the end shows arr[5] containing an unexpected value, a use-after-free demonstration with an empty Data field, and the expression 2147483647 + 1 wrapping to -2147483648.

Evidence limitation: the available qemu_gdb.log is only about 260 bytes and does not, in the captured tail, demonstrate a complete remote GDB session, breakpoint/watchpoint commands, backtrace, register inspection, or variable inspection at a crash. These required items should be added from a real GDB session before treating this section as complete.

Observed issue summary

Issue	Evidence available	Technical interpretation / proposed fix
Array bounds / unexpected value	The log shows arr[5] = 812441344 although arr[5] is outside the array's initialized memory. Keep all	Accessing outside the array's initialized memory. Keep all
Use-after-free demonstration	The log labels a use-after-free section; its Data field is empty in the captured output. Do not dereference stale pointer	Do not dereference stale pointer
Signed integer overflow	The log shows 2147483647 + 1 = -2147483648	Signed overflow in C is undefined behavior; validate bounds before addition

4. Section B — Memory Analysis with Valgrind

The recorded Valgrind report for the uncorrected application contains three error contexts and a definite memory leak. The fixed application report states that all heap blocks were freed and reports zero errors.

Metric	Before fix	After fix
Definitely lost	16,900 bytes in 250 blocks	0 bytes in use at exit
Indirectly lost	0 bytes in 0 blocks	0 bytes in use at exit
Total heap activity	See full log for allocation totals	352 allocs, 352 frees, 24,196 bytes allocated
Valgrind result	ERROR SUMMARY: 3 errors from 3 contexts	ERROR SUMMARY: 0 errors from 0 contexts

Interpretation: before the fix, Valgrind reported 16,900 definitely-lost bytes across 250 blocks, with three error contexts. After the fix, total allocations and frees match (352 each), no heap blocks remain at exit, and the error summary is zero. The full logs should accompany this report so the allocation stack traces can be reviewed.

The required before/after VmRSS measurement for the ARM target is not present in the currently confirmed evidence. Add the QEMU /proc/PID/status readings if available; do not infer VmRSS from the host Valgrind output.

5. Section C — strace and perf

The submission directory contains strace_full.txt, strace_filtered.txt, strace_summary.txt, perf_report.txt, perf_stat.txt, and perf.data. These are the evidence files for syscall tracing and CPU profiling. The summary outputs should be reviewed alongside the full logs to identify the most frequent or costly system calls and the functions with the highest sampled CPU cost.

Interpretation should use the actual values in the attached logs. This report does not claim specific syscall counts or CPU-hotspot percentages because those detailed values were not included in the terminal excerpt used to prepare this draft.

6. Section D — cyclicttest latency

The logs directory contains cyclicttest_baseline.txt, hist_standard.txt, and latency_histogram.png. The available previous run recorded thread 0: Min 26 μ s, Act 759 μ s, Avg 497 μ s, Max 6350 μ s; thread 1: Min 22 μ s, Act 347 μ s, Avg 510 μ s, Max 5278 μ s. These values should be interpreted as measurements from the recorded environment, which may be affected by virtualization and system load.

The histogram file only captures the available histogram range; the previously observed histogram data covers 0–199 μ s, so it does not represent the full long-latency tail up to the reported maxima. A PREEMPT_RT comparison run is not confirmed in the current file inventory, so this report does not claim a measured standard-versus-RT comparison.

7. QEMU target validation

The ARM Linux 5.15.0 target successfully booted under QEMU vexpress-a9 using initramfs_lab3.cpio.gz. The init script loaded lab2_driver successfully and created /dev/lab2 with major 240, minor 0. The /proc/lab2_info interface reported driver statistics, and sysfs exposed buffer_len, last_data, open_count, and dev. Writing 'LAB03 test' to /dev/lab2 and reading it back verified the device path and last_data reporting.

8. Conclusion

The available evidence demonstrates successful QEMU boot and device-driver validation, detection and correction of a Valgrind-reported memory leak, and the presence of strace, perf, and cyclicttest artifacts. Before final submission, the GDB session should be expanded to include the required debugging commands and outputs, and the report should be supplemented with the actual VmRSS measurements and a PREEMPT_RT comparison if required by the instructor. The current report explicitly identifies these evidence gaps rather than fabricating results.

Appendix — files included as evidence

apps/: buggy_app.c, buggy_app, leaky_app.c, leaky_app, leaky_app_fixed.c, leaky_app_fixed, perf_target.c, perf_target.
logs/: gdb_session.txt, valgrind_report.txt, valgrind_fixed.txt, strace outputs, perf outputs, cyclicttest outputs, histogram image, and environment_info.txt.
output/: initramfs_lab3.cpio.gz.